TECHIN 514 A

Second Device

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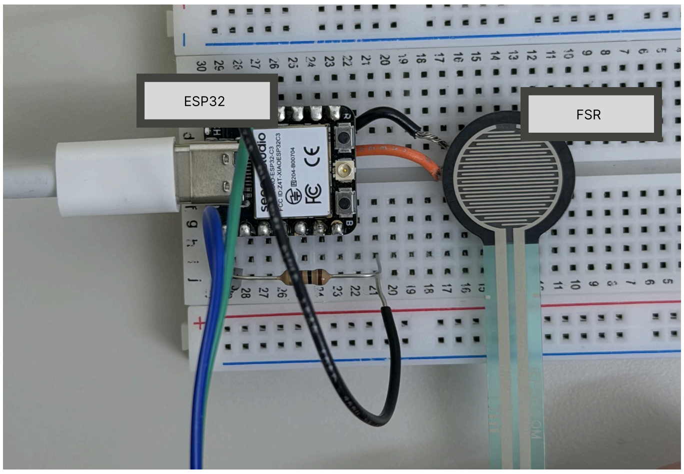
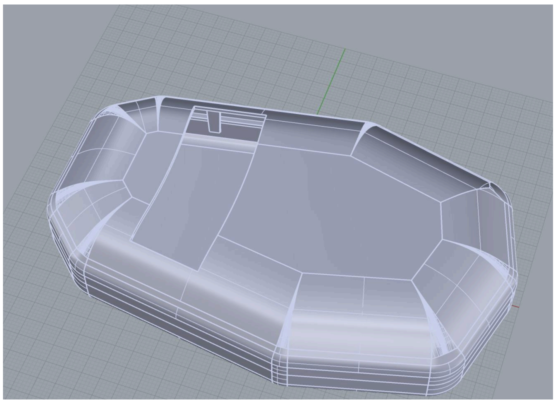
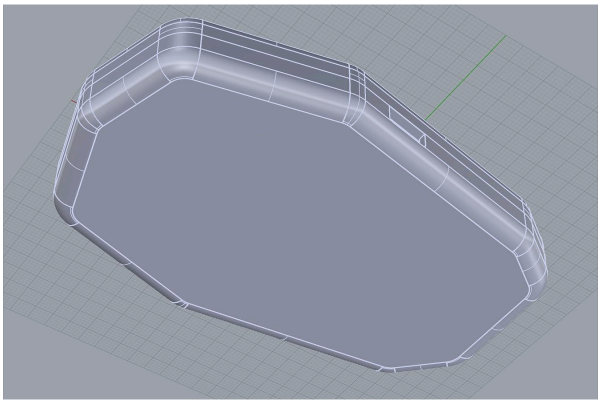
This sensor device uses an ESP32-C3 to read three FSR406 pressure sensors. It converts each channel to a voltage and prints the results to the Serial Monitor at a fixed sampling rate, ready to be forwarded wirelessly to the display device. The demo tests the code and connections using 1 sensor as example.

Photo of Functional Breadboard



Two screenshots of CAD Model





Two screenshots of CAD Model

This code reads a force-sensitive resistor (FSR) connected to GPIO pin 2 on a ESP32 using digital input. In the main loop, it checks every 500ms whether pressure is being applied to the sensor and prints "Pressure detected!" or "No pressure" to the serial monitor.

```
const int FSR_PIN = 2;
void setup() {
  Serial.begin(115200);
  pinMode(FSR_PIN, INPUT);
  Serial.println("FSR Sensor Ready");
}
void loop() {
  int state = digitalRead(FSR_PIN);
  if (state == HIGH) {
    Serial.println("Pressure detected!");
  } else {
    Serial.println("No pressure");
  }
  delay(500);
}
```

Github Repo

Under FinalProject/Code/ErgoCompass/src/main.cpp

https://github.com/Felix-Jr056/TECHIN514A-Wi26.git